

Practical 09 :

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code :

```
GNU nano 8.7.1
#include <stdio.h>

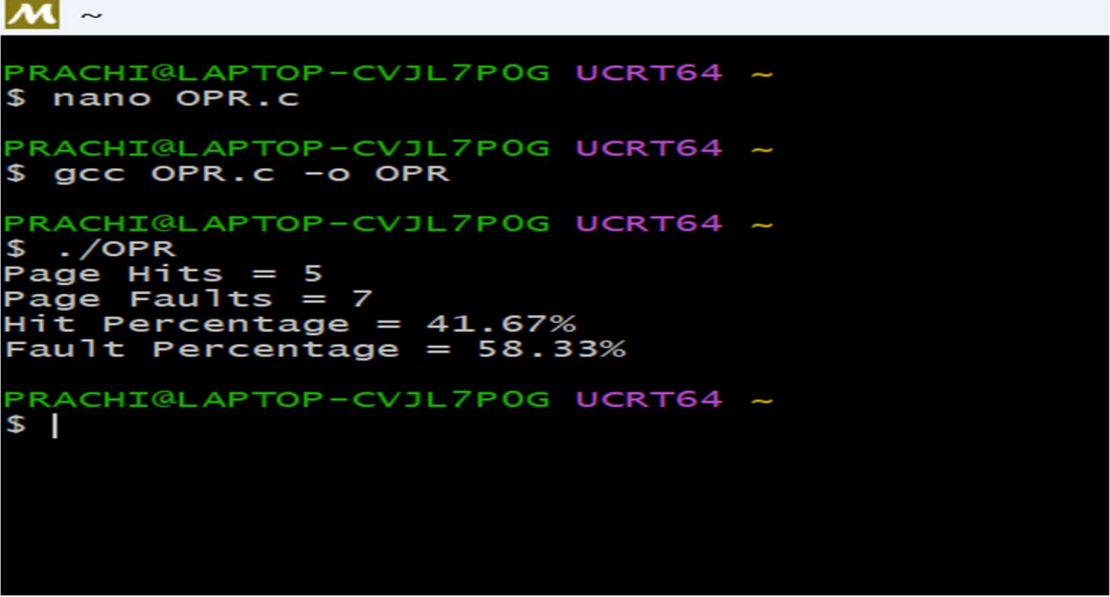
int main() {
    int pages[12] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[3];
    int faults = 0, hits = 0;
    int i,j,k,pos;

    for(i=0;i<3;i++){
        frame[i] = -1;
    }
    for(i=0;i<12;i++){
        int found = 0;
        for(j=0;j<3;j++){
            if(frame[j] == pages[i]){
                hits++;
                found = 1;
                break;
            }
        }
        if(!found){
            int farthest = -1;
            pos = -1;
            for(j=0;j<3;j++){
                int next = -1;
                for(k=i+1;k<12;k++){
                    if(frame[j] == pages[k]){
                        next = k;
                        break;
                    }
                }
                if(next == -1){
                    pos = j;
                    break;
                }
                if(next > farthest){
                    farthest = next;
                    pos = j;
                }
            }
            frame[pos] = pages[i];
            faults++;
        }
    }

    printf("Page Hits = %d\n", hits);
    printf("Page Faults = %d\n", faults);
    printf("Hit Percentage = %.2f%%\n", (hits/12.0)*100);
    printf("Fault Percentage = %.2f%%\n", (faults/12.0)*100);

    return 0;
}
```

Output :



```
PRACHI@LAPTOP-CVJL7P0G UCRT64 ~  
$ nano OPR.c  
PRACHI@LAPTOP-CVJL7P0G UCRT64 ~  
$ gcc OPR.c -o OPR  
PRACHI@LAPTOP-CVJL7P0G UCRT64 ~  
$ ./OPR  
Page Hits = 5  
Page Faults = 7  
Hit Percentage = 41.67%  
Fault Percentage = 58.33%  
PRACHI@LAPTOP-CVJL7P0G UCRT64 ~  
$ |
```